LAB 07

Signal Stage: a Library of Trend, Mean-Reversion, Momentum and Breakout Signals

Item	Detail
Topic	Topic 2: Technical Indicators, Trading Strategies and Metrics with TradingView and Alpaca
Pipeline stage	Signals
Duration	25 min
Simulator page	http://localhost:8000/#/backtest?strategy=ema_cross&symbols=SPY&fast=20&slow=50&costs=alpaca_retail
Mode	No coding. Runs ONLINE with your Alpaca paper keys (Integrations page), or OFFLINE on the demo data and the built-in paper broker. Paper trading only.

Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.

Scenario

Lion City Capital wants every strategy - from a TradingView prototype to an ML model - to produce the same thing: a target position per symbol, decided at the close and traded on the next bar. The quant team builds one signal library that the evaluation, search, execution, automation and diagnostic stages all call.

Objective

Pick and customise one strategy from each family and compare them after costs.

Before You Start

- Make sure the Trading Simulator is running and open http://localhost:8000 > Course Labs > Lab 07. It is installed once - see the guide 'Download, Install and Start the Trading Simulator Software' in the Learner Guide.
- Record every result you are asked to check - you will use it in the debrief and in the later labs.

Lab Steps - In the Trading Simulator

No coding is needed. Make sure the Trading Simulator is running (installed once - see the guide 'Download, Install and Start the Trading Simulator Software') and open http://localhost:8000. In the sidebar open Course Labs > Lab 07: the app shows the same steps with a Go button that opens each page pre-filled. Every step works offline on the bundled demo data and the built-in paper broker.

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Step 1. Trend: EMA 20/50 on SPY

Run the preset.

Open: http://localhost:8000/#/backtest?strategy=ema_cross&symbols=SPY&fast=20&slow=50&costs=alpaca_retail (Backtest SPY EMA 20/50, Alpaca costs)

Check: Note Sharpe and max drawdown vs buy-and-hold.

Step 2. Mean reversion: Bollinger on KO

Run Bollinger Band mean reversion on KO (window 20, k 2.0), then change k to 1.5 and run it again. Set Benchmark to KO so the buy-and-hold figures under each metric are KO's, not SPY's.

Open: http://localhost:8000/#/backtest?strategy=bollinger_reversion&symbols=KO&window=20&k=2 (Backtest KO Bollinger 20/2)

Check: Win rate is high, but losses are larger than wins - 'many small wins, rare big losses'.

Step 3. Momentum and breakout

On the Backtest page choose Time-series momentum, look-back 126, symbol QQQ, Benchmark QQQ, and run it. Then run the Donchian breakout on QQQ (entry 20, exit 10).

Open: http://localhost:8000/#/backtest?strategy=donchian_breakout&symbols=QQQ&entry=20&exit=10 (Backtest QQQ Donchian 20/10)

Check: Open the 'Price & trades' tab: buy arrows follow 20-day highs.

Step 4. Read each strategy's explanation

Strategy Library: open each card and read How it works / Rule / Works best when / Watch out for.

Open: http://localhost:8000/#/strategies

Check: You can say which family suits a trending market and which a range-bound one.

Expected in the app: Four backtests with tuned parameters and an explanation of when each family works.

Record your results in Lab07_Results.xlsx in this lab folder: the settings are filled in - type the metrics into the yellow cells. Its Definitions sheet explains every column.

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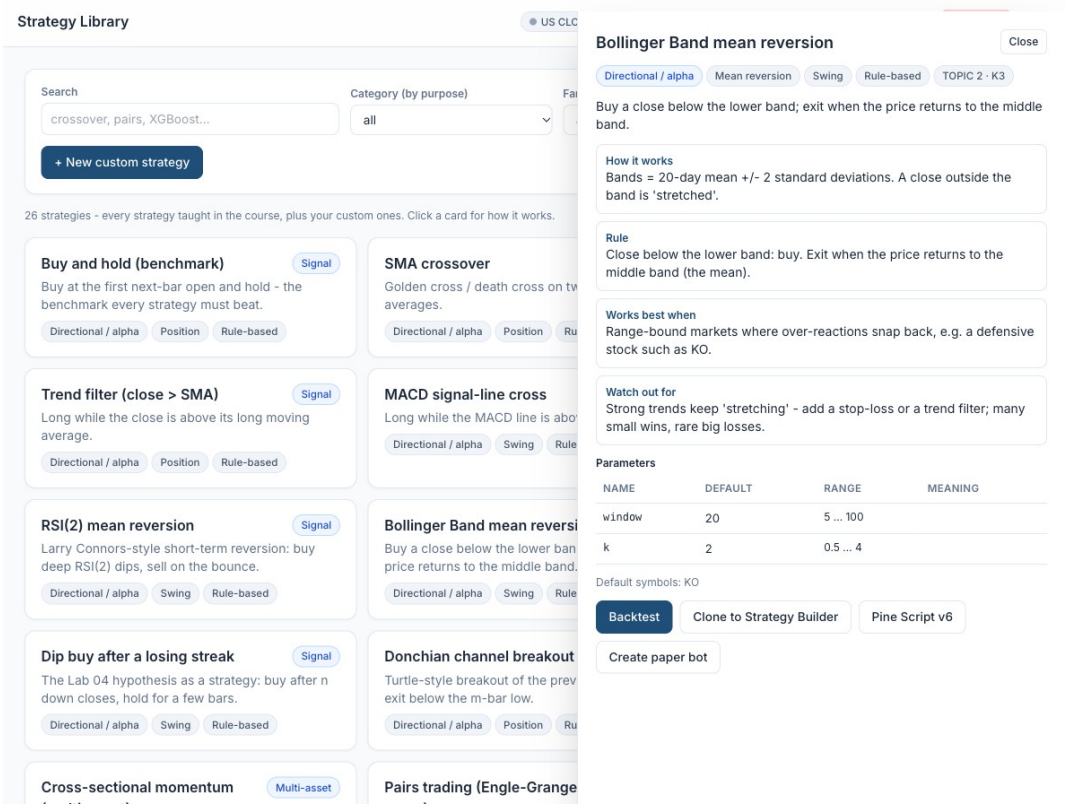


Figure: Lab 07 in the Trading Simulator.

Optional: Let AI Write the Code

This lab needs no coding. To see, run or customise the code behind it, click a link: the AI opens with the prompt filled in and writes the code for you. All prompts are also in labs/Prompt_Library.html and .pdf.

A library of four strategy families - run it in Google Colab
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: backtest four strategies on 2018-2025 daily data, trade at the next day's open with a 0.05% cost per trade: an EMA 20/50 crossover on SPY; Bollinger mean reversion on KO (buy below the lower band, exit at the middle band) with k = 2.0 and k = 1.5; 126-day time-series momentum on QQQ; and a Donchian 20/10 breakout on QQQ. Print CAGR, Sharpe ratio, maximum drawdown and number of trades in one table and explain when each family tends to work.

Open in Claude | Open in ChatGPT | Open Google Colab

Debrief

A single signal contract - target positions decided at the close, held from the next bar - is what lets one library feed backtests, live execution and diagnostics. Indicators become signals through simple, testable rules, and the number of entries each rule produces is the first estimate of its trading cost.

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